electrical contact. Thus a potential difference is developed across solar cell.When an external load is connected to the solar cell, photocurrent flows through the load.

Many solar cells are connected together either in series or in parallel combination to form a solar panel. Many solar panels are connected with each other to form solar arrays. For high power applications, solar panels and solar arrays are used.

Applications:

i) Solar cells are widely used in calculators, watches, toys, portable power supplies, etc.

ii) Solar cells are used in satellites and space applications.

iii) Solar panels are used for commercial production of electricity.

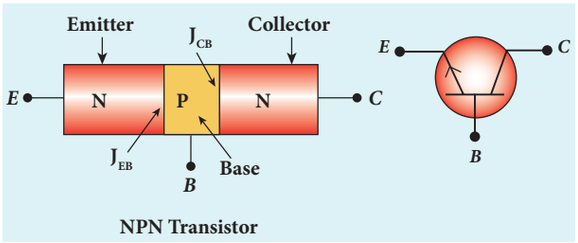
10.4 THE BIPOLAR JUNCTION TRANSISTOR [BJT]

Introduction

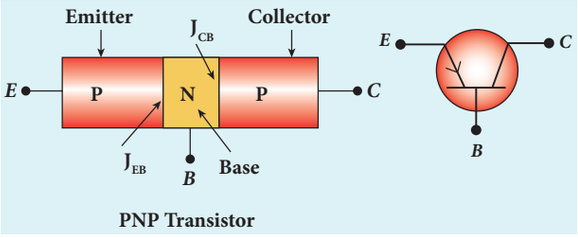
In 1951, William Schockley invented the modern version of transistor. It is a semiconductor device that led to a technological revolution in the twentieth century. The heat loss in transistor is very less. This has laid the foundation for integrated chips which contain thousands of miniaturized transistors. The emergence of the integrated chips led to increasing applications in the fast developing electronics industry.

Bipolar Junction Transistor (BJT)

The BJT consists of a semiconductor (silicon or germanium) crystal in which an *n*-type material is sandwiched between two *p*-type materials (PNP transistor) or a *p*-type material sandwiched between two *n*-type materials (NPN transistor). To protect it against moisture, it is sealed inside a metal or a plastic case. The two types of transistors with their circuit symbols are shown in Figure 10.25.



(a)



(b)

Figure 10.25 Schematic Diagram of (a) NPN transistor and circuit symbol (b) PNP transistor and circuit symbol

The three regions formed are called as emitter, base and collector which are provided with terminals or ohmic contacts labeled as *E, B*, and *C*. As BJT has two *p-n* junctions, two depletion layers are formed across the emitter-base junction (J_{EB}) and collector-base junction (J_{CB}) respectively. The circuit symbol carries an arrowhead at the emitter lead pointing from *p* to *n* indicating the direction of conventional current.

Emitter:

The main function of the emitter is to supply majority charge carriers to the collector region through the base region. Hence, emitter is more heavily doped than the other two regions.

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